

HFN 180/102
SAR. OHANGA MAXWELL

KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION SCIENCE
(HOME SCIENCE)

HFN 180 /102: INTRODUCTION TO FOOD AND NUTRITION

1ST SEMESTER 2025/2026

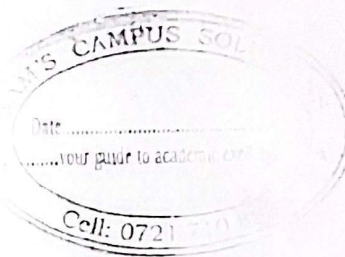
TIME: 2 HOURS

INSTRUCTIONS:

Answer ALL Questions in Section A and Section B

SECTION A (20 MARKS)

1. Define the following concepts:
 - (a) Nutrition (1 mark)
 - (b) Transamination (1 mark)
 - (c) Body Composition (1 mark)
 - (d) Deamination (1 mark)
2. Describe (6) six factors determining an individual's nutrient needs. (6 marks)
3. State the nutrient deficiencies that show the following disorders
 - (a) Pellagra (1 mark)
 - (b) Neuro-Tube Defects (1 mark)
 - (c) Osteomalacia (1 mark)
 - (d) Ariboflavinosis (1 mark)
4. Giving one example in each classify
 - (a) Carbohydrates into three (3 marks)
 - (b) Proteins based on quality (3 marks)



SECTION B (40 MARKS)

5. Using relevant examples, analyze five causes of protein Energy Malnutrition (PEM) among children below 5 years old in Kenya. (10 marks)
6. Giving examples, discuss five possible food based strategies to curb both macro and micronutrient deficiencies in Kenya. (10 marks)
7. Explain five healthy benefits of dietary fibre (10 marks)
8. Discuss five nutritional and health complications associated with obesity. (10 marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Page 1 of 1

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KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (FOOD,
NUTRITION AND DIETETICS)

HFN 102/180: INTRODUCTION TO FOOD AND NUTRITION

1ST SEMESTER 2021/2022

TIME: 2Hours

INSTRUCTIONS: Answer ALL questions in Section A and Section B

SECTION A: 30 Marks

1. Define the following concepts
 - (a) Nutrition (2 marks)
 - (b) Non-Essential Nutrients (2 marks)
 - (c) Basal Metabolic Rate (BMR) (2 marks)
 - (d) Estimated Average Requirement (EAR) (2 marks)
 - (e) Transamination (2 marks)
2. State the nutrient deficiencies that show the following disorders?
 - (a) Beri-beri (1 mark)
 - (b) Pellagra (1 mark)
 - (c) Neuro-Tube Defects (1 mark)
 - (d) Osteomalacia (1 mark)
 - (e) Ariboflavinosis (1 mark)
3. List five complications associated with obesity (5 marks)
4. Iron deficiency is a public health problem
 - (a) Describe five clinical signs of Iron deficiency (5 marks)
 - (b) Suggest five intervention measures for Iron deficiency (5 marks)

SECTION B: 40 Marks

5. Explain five healthy diet guidelines that you can recommend to a client (10 marks)
6. A high number of children below 5 years old in Kenya suffer from Protein Energy Malnutrition(PEM)
 - (a) Explain five causes of PEM in Kenya (10 marks)
 - (b) Differentiate between Marasmus and Kwashiorkor (10 marks)
7. Discuss five basic causes of malnutrition in the community (10 marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Page 1 of 1

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UNIVERSITY EXAMINATIONS 2010/2011

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
HFN 102: INTRODUCTION TO FOODS, NUTRITION AND DIETETICS

DATE: Monday 29th November, 2010

TIME: 8.00 a.m. – 10.00 a.m.

INSTRUCTIONS

Answer ALL questions.

1. a) The choice of diet influences long-term health, within the range set by genetic inheritance. Explain. [3 marks]
b) How does the body respond to starvation? [6 marks]
2. a) Briefly describe three characteristics of a nutritious diet. [6 marks]
b) Discuss three possible mechanisms by which fibre lowers serum cholesterol. [6 marks]
3. a) Explain the differences between saturated and unsaturated fatty acid. Give an example of each. [4 marks]
b) Describe four functions of lipids in the body. [4 marks]
4. a) Define the term oxidative stress. [4 marks]
b) Define 'retinal equivalent.' State two functions of vitamin A and explain respective deficiency disorders. [3 marks]
5. a) List six causes of vitamin D deficiency. [3 marks]
b) State two functions and deficiency symptoms of zinc. [4 marks]
6. a) Briefly explain two main factors that determine protein quality. [4 marks]
b) Explain two main classifications of proteins. Give an example of each. [6 marks]
7. a) Briefly describe three Dietary Reference Intake (DRI) values. [9 marks]
b) With the aid of a diagram, explain the Food Guide Pyramid and give 2 limitations. [8 marks]

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UNIVERSITY EXAMINATIONS 2010/2011

**MAY INTAKE SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE**

HFN 102: INTRODUCTION TO FOODS, NUTRITION AND DIETETICS

DATE: TUESDAY, 2ND AUGUST 2011

TIME: 11.00 A.M. – 1.00 P.M.

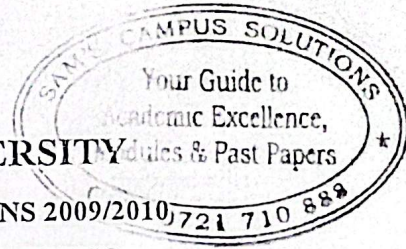
INSTRUCTIONS: ANSWER ALL QUESTIONS.

1. Explain in detail THREE functions of fibre in the diet. (9 marks)
2. Describe FOUR functions of water in the body. (12 marks)
3. Describe FOUR main functions of carbohydrates in the body and in the diet. (9 marks)
4. Describe the three functions of proteins in the body. (9 marks)
5. Explain the difference between
 - i) Essential amino acid and non-essential amino acid
 - ii) Saturated and non-saturated fat
 - iii) Protein of low biological value and high biological value(3 marks)
6. Explain THREE main functions of each of the following nutrients:
 - a) Vitamin C
 - b) Vitamin A
 - c) Calcium
 - d) Iron
 - e) Zinc(10 marks)
7. Explain what deficiencies in the following nutrients lead to and the symptoms of each:
 - i) Vitamin D
 - ii) Iodine
 - iii) Proteins
 - iv) Vitamin B₆(8 marks)
8. Describe FOUR functions of lipids in the diet and the body. (10 marks)

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Page 1 of 1

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UNIVERSITY EXAMINATIONS 2009/2010

INSTITUTE OF OPEN LEARNING

EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION

(HOME ECONOMICS)

HEN 102: INTRODUCTION TO FOODS, NUTRITION AND DIETETICS

DATE: SATURDAY, 20TH FEBRUARY 2010

TIME: 11.00 A.M. - 1.00 P.M.

INSTRUCTIONS: Answer ALL the questions.

1. Define the following terms:
 - a) Nutrition (2 marks)
 - b) Dietetics (2 marks)
 - c) Basal metabolic rate (2 marks)

2. Differentiate between the following terms:
 - a) Macronutrients and micronutrients (4 marks)
 - b) Essential and non essential amino acids (4 marks)
 - c) Saturated and non-saturated fats (4 marks)
 - d) Monosaccharide and polysaccharides (4 marks)

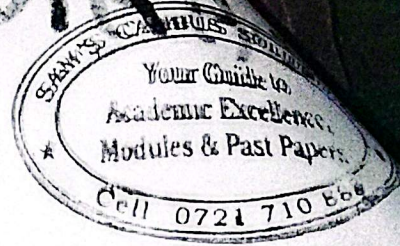
3. What is the role of the following nutrients in the body?
 - a) Vitamin D (2 marks)
 - b) Niacin (2 marks)
 - c) Sodium (2 marks)
 - d) Pyridoxine (2 marks)

4. What is the metabolic implication of the following:
 - a) Use of protein as a source of energy (4 marks)
 - b) Excessive lipid intake (4 marks)

Page 1 of 2

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ADMISSION

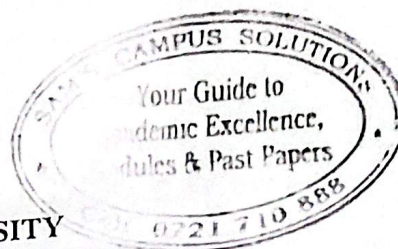


5. Describe how digestion takes place in a human body. (12 marks)
6. Discuss four major function of proteins in the body. (12 marks)
7. Describe the deficiency due to lack of the following nutrients.
 - a) Vitamin A (2 marks)
 - b) Iron (2 marks)
 - c) Zinc (2 marks)
 - d) Ascorbic acid (2 marks)

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KENYATTA UNIVERSITY



UNIVERSITY EXAMINATIONS 2017/2018
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
EDUCATION (HOME ECONOMICS)

HFN 102: INTRODUCTION TO FOOD NUTRITION AND DIETETICS

DATE: Monday 5th February 2018

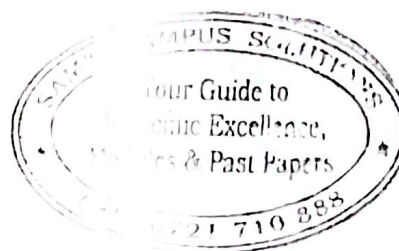
TIME: 4.30 p.m. – 6.30 p.m.

INSTRUCTIONS:

Answer ALL questions

1. State ONE best dietary source of the following nutrients (5 marks)

- (a) Iron
- (b) Vitamin A
- (c) Vitamin C
- (d) Vitamin B₁₂
- (e) Zinc



2. Briefly explain the relationship between the following

- (a) Calcium and vitamin D
- (b) Iron and vitamin C
- (c) Vitamin A and dietary fat
- (d) Calcium and phosphorous
- (e) Iron and zinc
- (f) Vitamin A and Health
- (g) Vitamin C and Health

(2 marks)
(2 marks)
(2 marks)
(2 marks)
(2 marks)
(5 marks)
(5 marks)

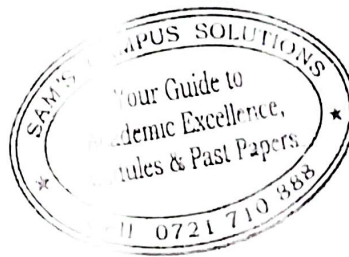
Page 1 of 2

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3. "Knowledge on nutrients is important for a nutritionist" based on that knowledge:
- (a) Describe the three monosaccharides (9 marks)
 - (b) Giving examples, describe the three categories of proteins depending on their quality (9 marks)
 - (c) Outline five health benefits of dietary fibre (5 marks)
 - (d) Advise on intake of lipids (4 marks)
4. Iron deficiency anaemia is of public health concern
- (a) State four causes ^{of iron} deficiency anaemia (4 marks)
 - (b) Explain five ways in which iron absorption can be enhanced (10 marks)
 - (c) Outline four intervention measures to iron deficiency anaemia (4 marks)



INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Page 2 of 2

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6. Explain 2 functions each of the following nutrients.

- a) Vitamin D
- b) Vitamin E
- c) Iron
- d) Zinc
- e) Iodine

[10 marks]

7. Explain what deficiencies in the following nutrients lead to

- a) Calcium
- b) Vitamin A
 - i) What are the symptoms of these deficiencies?
 - ii) How can they be corrected

[8 marks]

[6 marks]

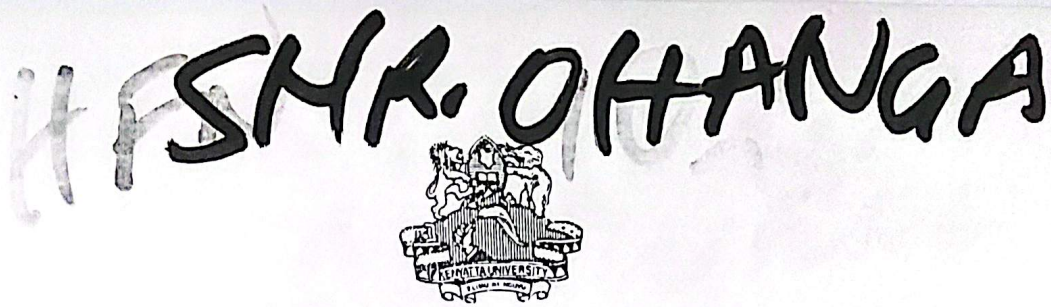
8. Explain three functions of water in the body

[6 marks]

9. Explain three functions of fibre in the diet

Page 3 of 3

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UNIVERSITY EXAMINATIONS 2010/2011

INSTITUTE OF OPEN LEARNING

**EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE AND
BACHELOR OF EDUCATION**

HFN 102: FOOD AND NUTRITION

DATE: WEDNESDAY 2ND FEBRUARY 2011

TIME: 8.00 A.M -10.00 A.M

INSTRUCTIONS:

Answer Section A in the question paper and Section B in the answer booklet provided.

Section A

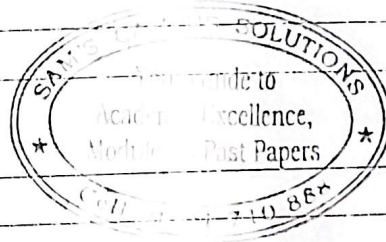
1. Define the following: (10 marks: 1 mark each)

a) Nutrient

b) Precursor

c) Gluconeogenesis

d) Deamination



Page 1 of 3

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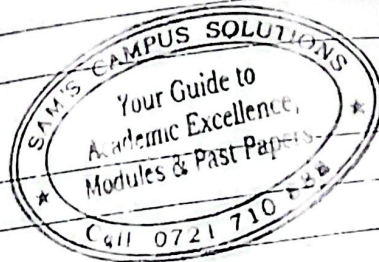
e) Hydrogenation

f) Basal Metabolism

g) Lactose intolerance

h) Recommended Dietary Allowance

i) RDA



SECTION B: Answer in the examination booklet provided

2. Describe four (4) main functions of proteins in the body. [10 marks]
3. Explain in detail the four (4) functions carbohydrates in the body. [10 marks]
4. Describe the three (3) functions lipids in the diet and in the body. [10 marks]
5. Explain the difference between
 - a) Essential and non-essential fatty acid
 - b) Saturated and non-saturated fat
 - c) Monosaccharide and disaccharide
 - d) Complete and incomplete protein[4 marks]

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KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2010/2011

OPEN, DISTANCE AND E-LEARNING EXAMINATION FOR THE DEGREE OF SCIENCE
(FOODS, NUTRITION AND DIETETICS)

HFN 102: INTRODUCTION TO FOODS, NUTRITION AND DIETETICS

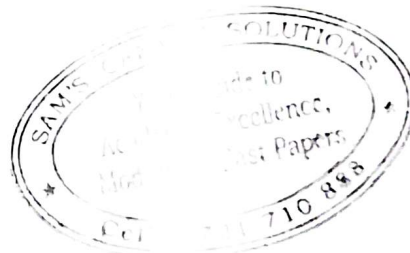
DATE: Thursday 7th July 2011

TIME: 4.30p.m – 6.30p.m

INSTRUCTIONS

ANSWER ALL QUESTIONS

1. Discuss in detail the four (4) functions of proteins in the body. [8marks]
2. Discuss four (4) main functions of carbohydrates in the body and in the diet. [12marks]
3. Using relevant examples, explain the meaning of the following:
 - a) Lactose intolerance
 - b) Precursor
 - c) Hydrogenation
 - d) Saponification
 - e) Buffer
 - f) Hydration
4. Explain the difference between.
 - i) Essential fatty acid and non-essential fatty acid
 - ii) Saturated and non-saturated fat
 - iii) Polysaccharide and Disaccharide
 - iv) Protein of high biological value and low biological value
5. Explain three (3) functions of each of the following nutrients. [8marks]
 - a) Vitamin A
 - b) Vitamin C
 - c) Iodine



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- d) Iron
- e) Zinc
- 6. Explain what deficiencies in the following nutrients lead to.
 - i) Vitamin D and Vitamin C
 - ii) State two (2) symptoms of each of the deficiencies
 - iii) How can each deficiency be corrected

[12marks]

- 7. Describe three (3) functions of fibre in the human body.
- 8. Describe two (2) functions of water.

[6marks]

[2marks]

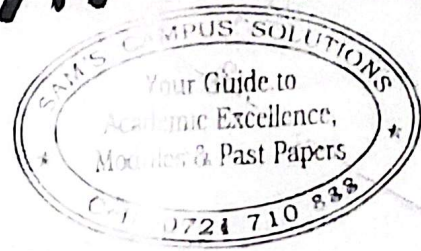


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KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2010/2011

INSTITUTIONAL BASED PROGRAMME (IBP) AUGUST SESSION
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (FOODS,
NUTRITION AND DIETETICS)

IFN 102: INTRODUCTION TO FOODS, NUTRITION AND DIETETICS

DATE: Thursday, 30th December, 2010

TIME: 11.00 a.m. – 1.00 p.m.

INSTRUCTIONS:

Answer ALL Questions

Q.1) Define the terms:

- a) Nutrients
- b) RDA

(2 marks)

Q.2) State FOUR (4) main functions of proteins in the body. (10 marks)

Q.3) Explain in details the FOUR (4) functions carbohydrates in the body. (10 marks)

Q.4) Describe the THREE (3) functions lipids in the diet and in the body. (10 marks)

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Page 1 of 2

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- Q.5) Explain the difference between
- i) Essential and non-essential fatty acid
 - ii) Saturated and non-saturated fat
 - iii) Monosaccharide and disaccharide
 - iv) Complete and incomplete protein

(8 marks)

- Q.6) Explain TWO (2) functions each of the following nutrients
- Vitamin D

Vitamin E

Iron

Zinc

Iodine

(10 marks)

- Q.7) a) Explain what deficiencies in the following nutrients lead to:

- i) Calcium
- ii) Vitamin A

b) What are the symptoms of these deficiencies

c) How can they be corrected?

(8 marks)

- Q.8) Explain THREE functions of water in the body

(6 marks)

- Q.9) Explain the function of fibre in the diet.

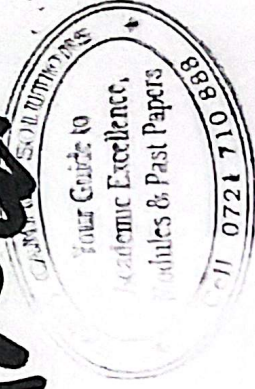
(6 marks)

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Page 2 of 2

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UNIVERSITY EXAMINATIONS 2009/2010

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCES (FOODS, NUTRITION AND DIETETICS)

HFN 102: INTRODUCTION TO FOODS, NUTRITION AND DIETETICS

DATE: THURSDAY 29TH JULY 2010 TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS

SECTION A: Answer ALL Questions on the Question paper for Section A.

1. Define the following:
Recommended dietary allowance

Oedema

Lipoprotein

Rickets

2. State two rich food sources of the following nutrients in Kenyan diets
(4 marks)

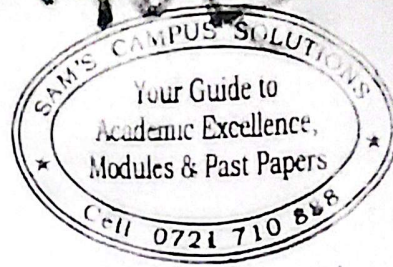
Rich Food Sources	
Nutrient	
Carbohydrates	
Proteins	
Lipids	
Vitamin A	
Vitamin D	
Vitamin C	
Iron	
Zinc	
Potassium	
Calcium	

(5 marks)

3. Give the structure of glucose (3 marks)

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4. i) Explain what the deficiencies in the following nutrients lead to:

Nutrient	What Deficiencies in nutrient lead to	Symptoms of the deficiencies
Iron		
Proteins		
Iodine		
Vitamin C		

(12 marks)

5. Differentiate between the following:
Essential Amino Acid and Non-essential amino acid

Saturated and non-saturated fatty acid

Monosaccharide and Polysaccharide

Protein of high biological value and proteins of low biological value

6. State six functions of water in the human body (4 marks)

7. State (4) four symptoms of marasmus (3 marks)

(2marks)

Page 2 of 3

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29th July 2010

Section B: Answer ALL questions in the Examination booklet provided

8. Explain in detail the three (3) functions carbohydrates in the body (9 marks)

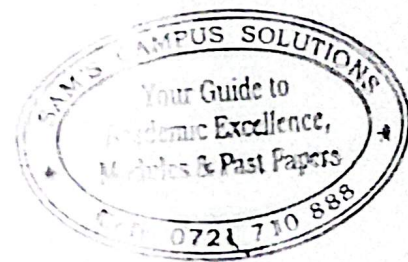
9. State three (3) main functions of proteins in the body (9 marks)

10. Describe the three (3) functions lipids in the diet and in the body (9 marks)

11. Explain 2 functions each of the following nutrients

Calcium
Vitamin D
Vitamin A
Iron
Zinc

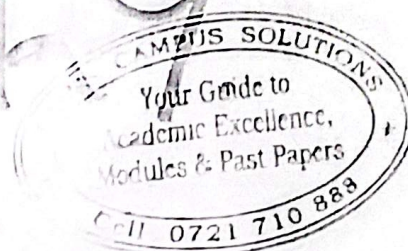
(10 marks)



Page 3 of 3

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DINO HENRY



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATION 2011/2012 ACADEMIC YEAR

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (FOODS, NUTRITION & DIETITICS)

HFN 102: INTRODUCTION TO FOODS, NUTRITION AND DIETETICS

DATE: Tuesday, 15th December, 2011

TIME: 8.00a.m. – 10.00a.m.

INSTRUCTIONS:

Answer ALL Questions

SECTION A: [40 Marks]

1. Define the following terms

(a) Metabolism.	[1 Mark]	
(b) Nutritional Status.	[1 Mark]	
(c) Kilocalorie.	[1 Mark]	
(d) Basal metabolic rate.	[1 Mark]	
(e) Teratogenicity.	[1 Mark]	
(f) Cretinism.	[1 Mark]	

2. Distinguish between the following terms.

(a) Glycolysis and gluconeogenesis.	[2 Marks]	
(b) Soluble and insoluble fibres.	[2 Marks]	
(c) Complete and incomplete proteins.	[2 Marks]	
(d) Saturated and unsaturated fatty acids.	[2 Marks]	
(e) Deamination and decalcification.	[2 Marks]	
(f) Sucrose and Cellulose.	[2 Marks]	

Page 1 of 2

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3. Outline two principles of diet planning. [2 Marks]
4. Explain why proteins are considered a wasteful source of energy. [2 Marks]
5. Explain why glucose is referred to as a blood sugar. [2 Marks]
6. Explain two functions of dietary fibre in the diet. [6 Marks]
7. Explain two main functions of the following nutrients [10 Marks]
- (a) Vitamin A
 - (b) Iron
 - (c) Iodine
 - (d) Vitamin B6
 - (e) Proteins

SECTION B: [30 Marks]

8. Briefly describe the five factors that influence food choice. [5 Marks]
9. Briefly explain five ways in which nutrition assessments are carried out among individuals. [5 Marks]
10. Water has important functions in the human body, explain. [5 Marks]
11. Describe the relationship between Diet, health and chronic diseases. [5 Marks]
12. Explain giving detailed examples five functions of carbohydrates in the body. [10 Marks]

